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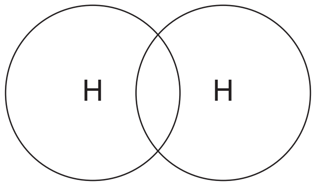
5. Hydrogen could be used to produce hydrogen chloride gas.

(a) The diagram represents the bonding in hydrogen. Complete the dot and cross diagram for a molecule of hydrogen. [1]



(b) The reaction between hydrogen and chlorine can be represented by the following equation.

$$H-H \quad + \quad Cl-Cl \longrightarrow H-Cl \quad + \quad H-Cl$$

The bond energy values are shown in the table.

Bond	Energy needed to break or form the bond (kJ)
H–H	436
Cl–Cl	242
H–Cl	431

Use the information in the table to answer the following questions.

(i) Calculate the energy needed to break all the bonds in the reactants. [2]

energy = ... kJ

(ii) Calculate the energy given out when all the bonds in the product are formed. [2]

energy = ... kJ

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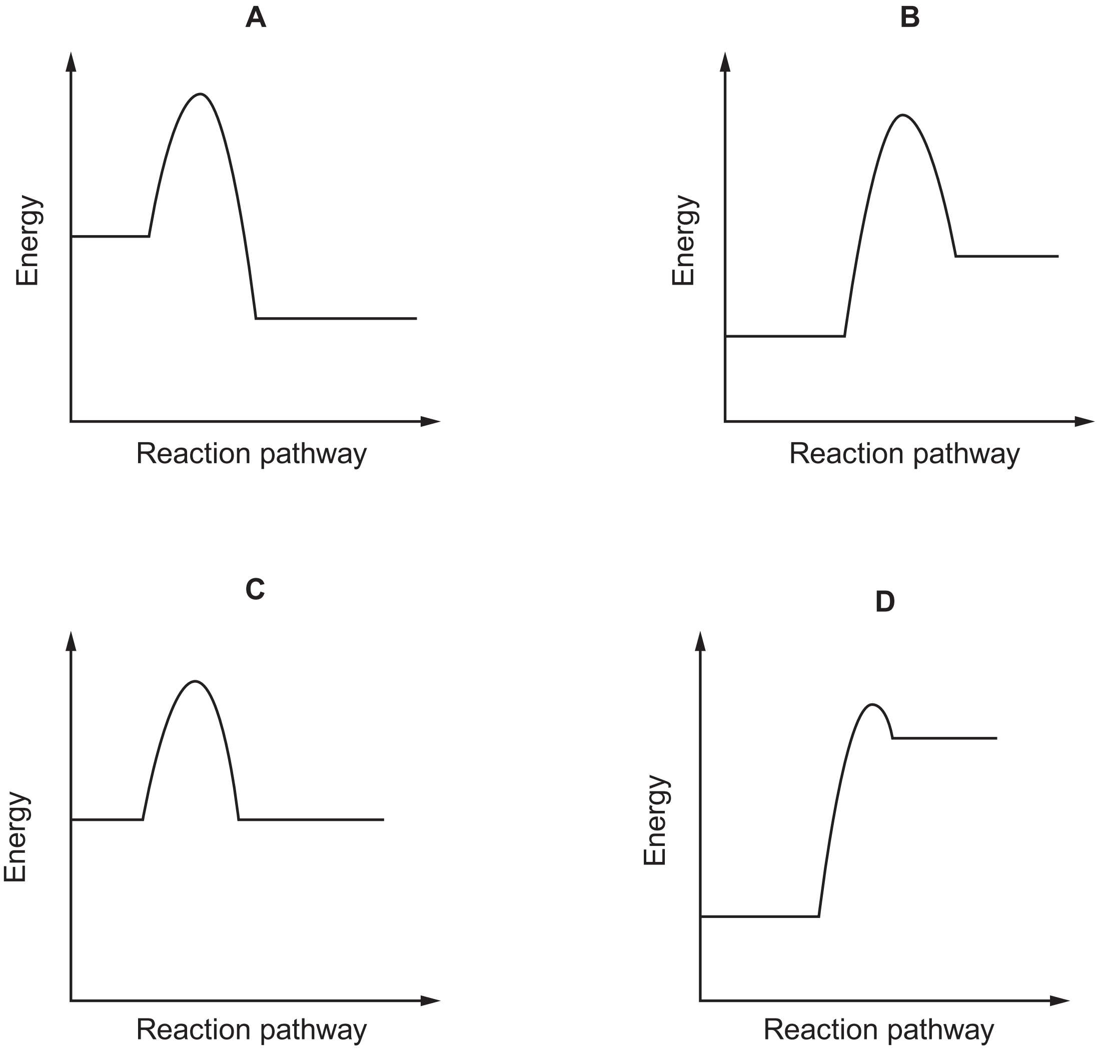
(iii) Possible energy diagrams for this reaction are shown below.



I. Select the correct energy diagram for this reaction. .... [1]

II. Explain your choice. [2]

....

....

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(c) Use words from the box to complete the sentences below about an **endothermic** reaction. [4]

more	less	taken in	given out	increase	decrease	thermal runaway

In an endothermic reaction .... energy is stored in the bonds **after** the reaction than before. This means energy is .... from the surroundings.

This will cause the temperature to .... during the reaction.

This cannot lead to a .... reaction.

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